

Harness State Engineering for Reliable Agentic AI Systems

A Runtime-State-Centric Survey Draft

ANONYMOUS FOR SUBMISSION,

Large language model (LLM) agents are increasingly deployed in long-horizon, tool-using, and multi-step environments where reliability depends not only on the underlying model but also on the execution harness surrounding it. Recent surveys have established agent harness engineering as an independent systems layer and have proposed harness taxonomies spanning execution, tooling, context, lifecycle, observability, verification, and governance. In parallel, recent work has shown that coding-agent harnesses can be automatically evolved through observability-driven loops, while trajectory-level safety research has demonstrated that output-level evaluation alone is insufficient for judging harness correctness. Motivated by these developments, this survey proposes Harness State Engineering as a complementary runtime-state-centric perspective for studying reliable agentic systems. Whereas context engineering focuses on what information the model should see at inference time, Harness State Engineering focuses on what the runtime must track, control, validate, recover, and audit during execution. We organize the literature around seven state dimensions—execution, tool, memory/context, verification, observability, governance, and recovery state—and use this taxonomy to synthesize findings across recent harness research.

1 Introduction

The recent shift from single-turn prompting toward persistent, tool-using, and multi-agent execution has changed where the main engineering bottleneck lies in AI systems. Context engineering formalizes the problem of optimizing the information payload assembled at inference time, including retrieval, processing, memory, and tool context [4]. By 2026, harness-oriented surveys and empirical papers increasingly argue that real-world reliability depends not only on what the model sees, but also on the infrastructure layer that wraps the model during execution [1, 5].

This survey is motivated by a gap between these two perspectives. Context engineering provides a strong conceptual framework for the input side of agent behavior. Harness engineering, by contrast, has emerged as a systems discipline for the execution side of agent behavior: how agents are orchestrated, constrained, instrumented, validated, and governed in production [1, 5]. Existing harness literature has made major progress by formalizing harness architectures, introducing observability-driven harness evolution, and auditing trajectory-level safety [2, 3]. However, a systematic survey that treats runtime state itself as the primary object of harness engineering remains underdeveloped.

We therefore propose **Harness State Engineering** as a complementary perspective: if context engineering asks what the model should see, Harness State Engineering asks what the runtime must know, store, update, validate, and audit while the agent acts.

2 Contributions

- We introduce Harness State Engineering as a runtime-state-centric lens for studying reliable agentic AI systems.
- We synthesize the literature into seven state dimensions: execution, tool, memory/context, verification, observability, governance, and recovery state.
- We clarify the conceptual distinction between context engineering and harness state engineering, showing how the two are complementary rather than competing.
- We connect recent work on harness taxonomies, automatic harness evolution, and trajectory-level safety auditing under a common runtime-state perspective.

- We identify open research problems in formal semantics, multi-agent state coordination, auditable harness evolution, and regulated enterprise deployment.

3 Related Work and Gap

Recent survey literature already provides strong foundations for harness engineering. One survey formalizes the harness as a six-component tuple $H = (E, T, C, S, L, V)$ [5]. Another survey proposes the ETCLOVG taxonomy, organizing harness engineering into execution environment, tool interface, context management, lifecycle/orchestration, observability, verification, and governance [1]. Beyond surveys, AHE shows that coding-agent harnesses can be automatically evolved via matched observability loops [2]. HarnessAudit shows that final answers alone do not reveal safety failures during execution and argues for full-trajectory auditing [3].

These works still leave space for a survey organized around runtime state as the central abstraction. Existing surveys emphasize layers, systems, and ecosystem coverage. AHE emphasizes harness evolution. HarnessAudit emphasizes safety auditing. What remains less systematically organized is the question of what runtime state a harness must maintain to support control, validation, recovery, and auditability across long-running agentic workflows.

4 A Formal View of Harness State

We define harness state at time t as

$$H_s(t) = (E_t, T_t, M_t, V_t, O_t, G_t, R_t)$$

where E is execution state, T tool state, M memory/context state, V verification state, O observability state, G governance state, and R recovery state.

5 The Seven State Dimensions

5.1 Execution State

Execution state captures where the agent is in its workflow, what step is active, which subtask is pending, and whether the current run is resumable after interruption.

5.2 Tool State

Tool state captures the runtime status of external tools and interfaces: availability, permissions, schema versions, cached outputs, error conditions, and calls already made.

5.3 Memory and Context State

Memory/context state records what information has been retrieved, compressed, persisted, or discarded across turns. It includes short-term working state, long-term memory references, retrieval provenance, and policies for context compaction or expiry.

5.4 Verification State

Verification state captures whether intermediate and final actions satisfy expected constraints. It may include schema checks, policy checks, guardrail decisions, milestone completion, test outcomes, or model-graded evaluations.

5.5 Observability State

Observability state captures traces, logs, metrics, failure classifications, and cross-component correlations.

5.6 Governance State

Governance state records the policy boundaries under which the agent is allowed to act, including approvals, role bindings, and audit entities.

5.7 Recovery State

Recovery state records how the system should respond after failure, interruption, regression, or detected policy risk.

6 Conclusion

Harness engineering has quickly become central to reliable agentic AI systems, but the next conceptual step is to study not only harness architecture, but also the runtime state that makes controlled execution possible. This survey draft proposes Harness State Engineering as that next step.

References

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